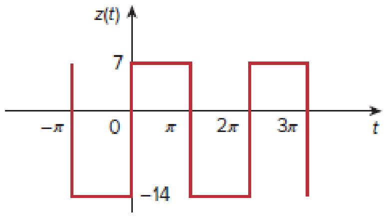
Problem

Obtain the Fourier series expansion for the waveform shown in Fig. 17.49.

Figure 17.49:



(a)

Step-by-step solution

Step 1 of 5

Refer to Figure 17.49 in the textbook for a waveform.

Write the mathematical representation of the waveform.

$$f(t)=\begin{cases}7, & 0<t<\pi,\\ -14, & \pi<t<2\pi.\end{cases}$$

The time period of the waveform is,

$$T=2\pi$$

Therefore, the fundamental frequency, ω_0 is,

$$\begin{aligned}\omega_0&=\frac{2\pi}{T}\\&=\frac{2\pi}{2\pi}\\&=1\end{aligned}$$

The trigonometric Fourier series of the function is,

$$f(t)=a_0+\sum_{n=1}^{\infty}\left[a_n\cos n\omega_0 t+b_n\sin n\omega_0 t\right]$$

Here,

$$a_0=\frac{1}{T}\int_0^T f(t)\,dt$$

$$a_n=\frac{2}{T}\int_0^T f(t)\cos(n\omega_0 t)\,dt$$

$$b_n=\frac{2}{T}\int_0^T f(t)\sin(n\omega_0 t)\,dt$$

Step 2 of 5

Calculate the value of a_0.

$$\begin{aligned}a_0 &= \frac{1}{T}\int_0^T f(t)\,dt \\ &= \frac{1}{2\pi}\left\{\int_0^{\pi} 7dt + \int_{\pi}^{2\pi} -14dt\right\} \\ &= \frac{1}{2\pi}\left\{7[t]_0^{\pi} - 14[t]_{\pi}^{2\pi}\right\} \\ &= \frac{1}{2\pi}[7\pi - 14\pi]\end{aligned}$$

Simplify the expression.

$$\begin{aligned}a_0 &= \frac{-7\pi}{2\pi} \\ &= -\frac{7}{2}\end{aligned}$$

Thus, the value of a_0 is $\boxed{-\frac{7}{2}}$.

Step 3 of 5

Calculate the value of a_n.

$$\begin{aligned}a_n &= \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\,dt \\ &= \frac{2}{2\pi}\left\{\int_0^{\pi} 7\cos nt\,dt + \int_{\pi}^{2\pi} -14\cos nt\,dt\right\} \\ &= \frac{1}{\pi}\left\{7\left[\frac{\sin nt}{n}\right]_0^{\pi} + 14\left[\frac{-\sin nt}{n}\right]_{\pi}^{2\pi}\right\} \\ &= \frac{7}{\pi}\left\{\left[\frac{\sin n\pi - 0}{n}\right] - 2\left[\frac{\sin 2n\pi - \sin n\pi}{n}\right]\right\}\end{aligned}$$

Simplify the expression.

$$a_n = 0 \quad (\text{since, } \sin n\pi = 0)$$

Thus, the value of a_n is $\boxed{0}$.

Step 4 of 5

Calculate the value of b_n.

$$\begin{aligned} b_n &= \frac{2}{T}\int_0^T f(t)\sin(n\omega_0 t)\,dt \\ &= \frac{2}{2\pi}\left\{\int_0^{\pi} 7\sin nt\,dt + \int_{\pi}^{2\pi} -14\sin nt\,dt\right\} \\ &= \frac{1}{\pi}\left\{7\left[\frac{-\cos nt}{n}\right]_0^{\pi} - 14\left[\frac{-\cos nt}{n}\right]_{\pi}^{2\pi}\right\} \end{aligned}$$

Simplify the expression.

$$\begin{aligned} b_n &= \frac{7}{\pi}\left\{\left[\frac{-\cos n\pi + 1}{n}\right] + 2\left[\frac{\cos 2n\pi - \cos n\pi}{n}\right]\right\} \\ &= \frac{7}{n\pi}\left\{\left[-(-1)^n + 1\right] + 2\left[1 - (-1)^n\right]\right\} \\ &= \frac{7}{n\pi}\left\{3 - 3(-1)^n\right\} \\ &= \frac{21}{n\pi}\left\{1 - (-1)^n\right\} \end{aligned}$$

Thus, the value of b_n is $\boxed{\frac{21}{n\pi}\left[1 - (-1)^n\right]}$.

Step 5 of 5

The Fourier series expansion is,

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\right]$$

Substitute the values of a_0, a_n, and b_n.

$$\begin{aligned} f(t) &= -\frac{7}{2} + \sum_{n=1}^{\infty}\left[0 + \frac{21}{n\pi}\left[1 - (-1)^n\right]\sin nt\right] \\ &= -3.5 + \frac{21}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\left[1 - (-1)^n\right]\sin nt \end{aligned}$$

Thus, the Fourier series expansion of the function $f(t)$ is,

$$\boxed{-3.5 + \frac{21}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\left[1 - (-1)^n\right]\sin nt}$$